

EXP 32: Prolog Program to Implement Pattern Matching

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Aim:

To develop a Prolog program that uses pattern matching (unification) to classify a triangle as equilateral, isosceles or scalene based on its side lengths.

Algorithm:

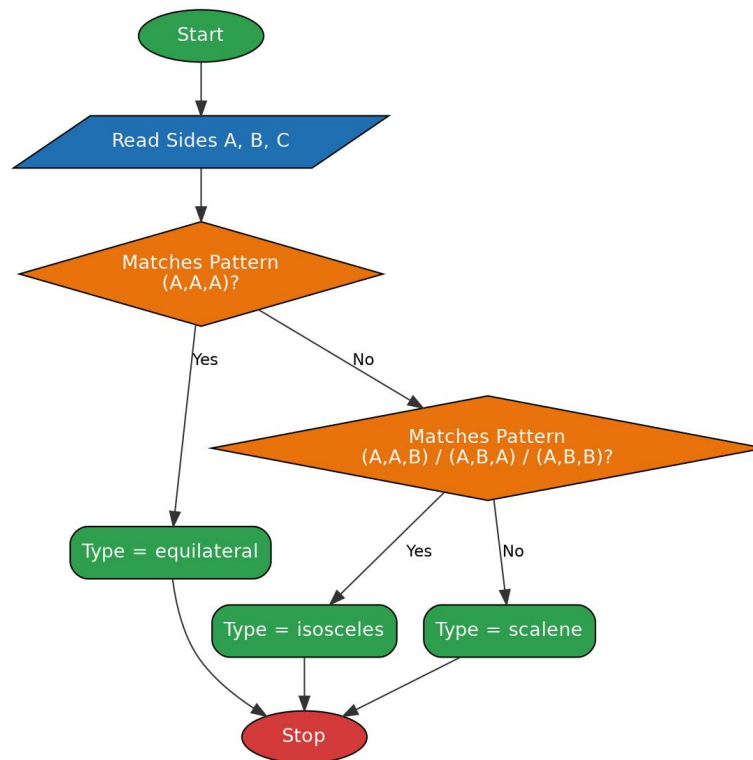
- Step 1: Start the program.
- Step 2: Accept the lengths of three sides of a triangle.
- Step 3: Compare the side lengths using pattern matching.
- Step 4: If all sides are equal, classify it as equilateral.
- Step 5: If any two sides are equal, classify it as isosceles.
- Step 6: If all sides are different, classify it as scalene.
- Step 7: Display the identified triangle type.
- Step 8: Verify the result.
- Step 9: Stop the program.

Source Code:

% Program to implement Pattern Matching (Triangle Classification)

```
triangle_type(A, A, A, equilateral) :- !.  
triangle_type(A, A, B, isosceles) :- A \= B, !.  
triangle_type(A, B, A, isosceles) :- A \= B, !.  
triangle_type(A, B, B, isosceles) :- A \= B, !.  
triangle_type(_, _, _, scalene).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [triangle].
true.

?- triangle_type(5,5,5,T).
T = equilateral.

?- triangle_type(5,5,3,T).
T = isosceles.

?- triangle_type(4,5,6,T).
T = scalene.
```

Result:

Thus the Prolog program to implement pattern matching was written, executed successfully and the output was verified.